Apr. 23 Revisiting last proof from previous class

n is composite

- Let D be an integral domain of characteristic n. By way of contradiction, suppose $n = st$ with $1 < s, t < n$. We have $0 = \underbrace{1+\dots+1}_{n \text{ times}} = \underbrace{\underbrace{(1+\dots+1)}_{s \text{ times}} + \dots + \underbrace{(1+\dots+1)}_{s \text{ times}}}_{t \text{ times}}$. Write $\sigma = \underbrace{1+\dots+1}_{s \text{ times}}$ and $\tau = \underbrace{1+\dots+1}_{t \text{ times}}$.

We have $0 = \underbrace{\sigma+\dots+\sigma}_{t \text{ times}} = \underbrace{1\cdot\sigma+\dots+1\cdot\sigma}_{t \text{ times}} = \underbrace{(1+\dots+1)}_{t \text{ times}}\cdot\sigma = \tau\sigma$. Thus $\tau = 0$ or $\sigma = 0$, which contradicts the supposition that the characteristic is n. (either t or s would be the char, since $\tau = 0 \Rightarrow \underbrace{1+\dots+1}_{t \text{ times}} = 0$, $\sigma = 0 \Rightarrow \underbrace{1+\dots+1}_{s \text{ times}} = 0$) $\square$

because D is an integral domain

- Note: prime characteristic $\not\Rightarrow$ integral domain

ex. $\mathbb{F}_2 \times \mathbb{F}_2$ has characteristic 2, but has zero divisors $\mathbb{F}_2 \cong \mathbb{Z}/2\mathbb{Z}$ $(\bar{1},\bar{0})\cdot(\bar{0},\bar{1}) = (\bar{0},\bar{0})$

What we've done, where we're going

- Groups
  - Properties (order, cyclic, etc)
  - Subgroups
  - Normal subgroups
  - Quotient/factor groups
  - Group homomorphisms
  - 1st Isomorphism Thm.
- Rings
  - Properties (characteristic, integral domain, etc)
  - Subrings
  - Ideals
  - Quotient/factor rings
  - Ring homomorphisms
  - 1st Isomorphism Thm.

★ We will only talk about ideals for commutative rings, otherwise have to be careful with left and right ideals

- Def. A subring I of a ring R is called a two-sided ideal of R if $\forall r \in R$ and $\forall a \in I$, both ra and ar are in I

- Def. If R is a ring and I is an ideal, the set of cosets $R/I = \{r+I : r \in R\}$ forms a ring with $(r+I)+(s+I) = (r+s)+I$ and $(r+I)(s+I) = rs+I$ $\forall r,s \in R$. This ring is called the quotient of R by I.

- Def. A function $\phi: R \to S$ (rings) is a homomorphism if $\phi(a+b) = \phi(a)+\phi(b)$ and $\phi(ab) = \phi(a)\phi(b)$ $\forall a,b \in R$. It is an isomorphism if it is also bijective.

- 1st (Ring) Isomorphism Thm. If $\phi: R \to S$ is a homomorphism of rings, then $R/\ker\phi \cong \phi(R)$, where $\ker\phi = \{r \in R : \phi(r) = 0\}$ and $\phi(R) = \{s \in S : \phi(r) = s \text{ for some } r \in R\}$.

Ch. 14: Ideals and Quotient rings

- **Def.** A subring I in a ring R is a (two-sided) ideal of R if $\forall r \in R$ and $\forall a \in I$ the products ra and ar are in I. Multiplicative Absorption property
  - The whole ring is always an ideal of itself due to closure, so we say an ideal I of R is proper if $I \subsetneq R$ (i.e $I \subseteq R$ but $I \neq R$)

Examples

- $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$
- $\{\bar{2}, \bar{4}, \bar{6}, \bar{8}, \bar{0}\}$ is an ideal of $\mathbb{Z}/10\mathbb{Z}$
- $\mathbb{Z}[i] = \{a+bi : a, b \in \mathbb{Z} \text{ and } i^2 = -1\}$ Gaussian integers, "$\mathbb{Z}$ adjoined i", Let $I = \{\alpha \in \mathbb{Z}[i] : \alpha = \beta \cdot (1-i) \text{ for some } \beta \in \mathbb{Z}[i]\}$. It turns out I is an ideal of $\mathbb{Z}[i]$

- Important example: Let R be a commutative ring. The set $\langle a \rangle \overset{\text{def}}{=} \{ra : r \in R\}$ (multiples of a) is an ideal of R. It is called the principal ideal generated by a.
  ↳ generated by one element

- **Thm. 14.1** (the Ideal Test) A nonempty subset I of a ring R is an ideal if
  1) $a - b \in I \ \forall a, b \in I$ (I is a subgroup of $(R, +)$)
  2) ra and ar are in $I \ \forall r \in R$ and $\forall a \in I$ (mult. absorption property, stronger than closure under mult.)

  Proof Suppose $I \subseteq R$ is a nonempty subset that satisfies properties 1 and 2.
  We see $a - b \in I$ and $ab \in I$ (inherited from R, a, b both in I) $\forall a, b \in I$. Thus I is a subring of R.
  Since 2 is exactly the mult. absorption property and since I is a subring of R, we see I is an ideal of R. $\square$

☆ Subgroup generated by $a \subseteq$ principal ideal generated by a

Ex. $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$.
- Clearly, $n\mathbb{Z}$ is nonempty.
- If $a, b \in \mathbb{Z}$, we see $a = nk$ and $b = nm$ for some $k, m \in \mathbb{Z}$. Thus $a - b = mk - nm = n(k-m) \in n\mathbb{Z}$.
- If $r \in \mathbb{Z}$, then $ra = ar = nkr \in \mathbb{Z}$.

Thus, the ideal test shows $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

- Claim: $\langle a \rangle$ (principal ideal generated by a) is an ideal of R

  Proof Let $x, y \in \langle a \rangle$. We see $x = a \cdot s$ and $y = a \cdot t$ for some $s, t \in R$. Thus $x - y = as - at \overset{\text{distributivity of ring}}{=} a(s-t) \in \langle a \rangle$.
  Let $r \in R$. We see $ra = ar \in \langle a \rangle = \{ar : r \in R\}$. Thus, the ideal test shows $\langle a \rangle$ is an ideal of R. $\square$